(Please write your Enrollment Number) Enrollment No. 08201032024

MID-TERM EXAMINATION

Course Name: B.Tech (CSEAI, ECEAI, IT, AI&ML, Semester: 3rd

(October, 2025) OFF LINE mode

Subject Code: BIT 201	Subject: Data Structures and Algorithm
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Write a function to return the second largest element from a sorted array.	

	b) What is the time complexity for the following code segment
	Void func(int n){
	For(int i = 0; i < n ; i=i+2){
	Cout<<" "<<i;
	}}	

	c) Convert the expression to prefix using stacks (A*B+C/D-E)	
	d) Write a function to return 'n'th element from a doubly linked list.	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) For a 2D array of 5 x 10 elements stored at location starting at 1000, having 2 bytes as the size of each element. What is the row major and column major addresses for the element arr[3][8].	
	b) Write a function to check whether a given string is palindrome or not?	
	c) For an array of N elements, how many recursive calls will the binary-search algorithm take? Justify.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Implement Singular queue using arrays (insert() and delete() operations), what is the problem in this implementation and how can it be resolved?	
	b) Write a program to reverse a circular linked list.	
	c) Convert the following expression to postfix and evaluate using stacks. (5+2)*(8-6)/(2+4/2-3)	